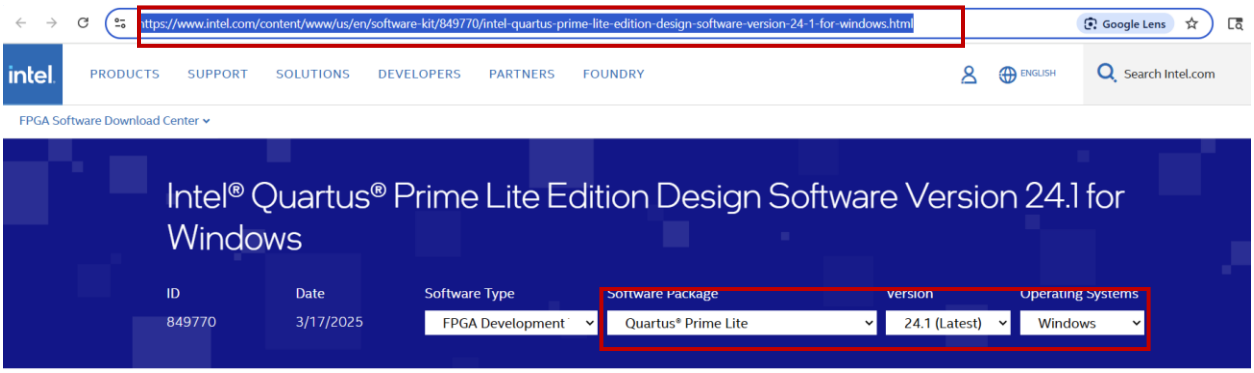
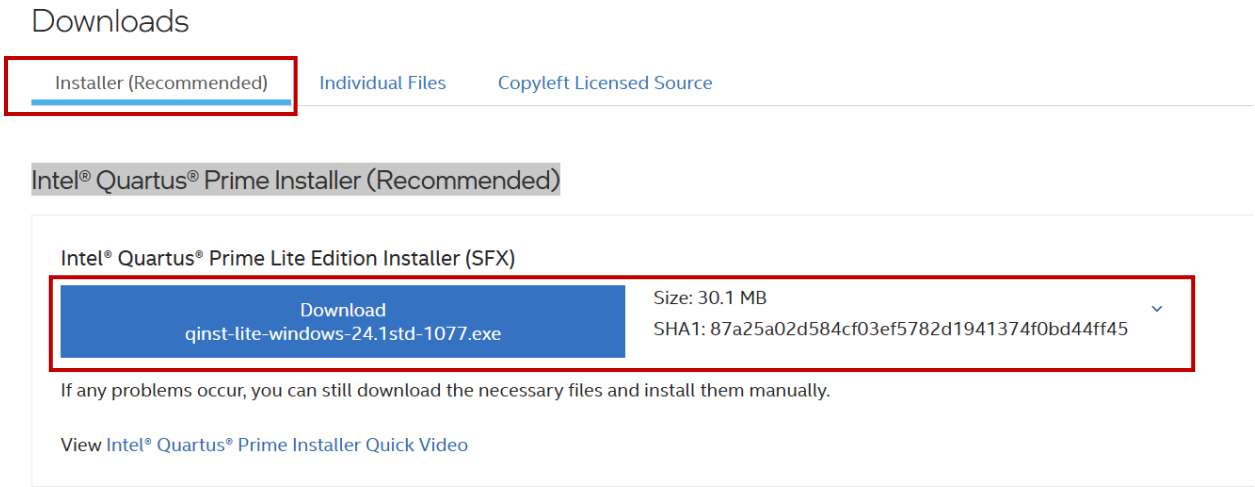
Installing quartus software (Follow the snapshots and red squares!)

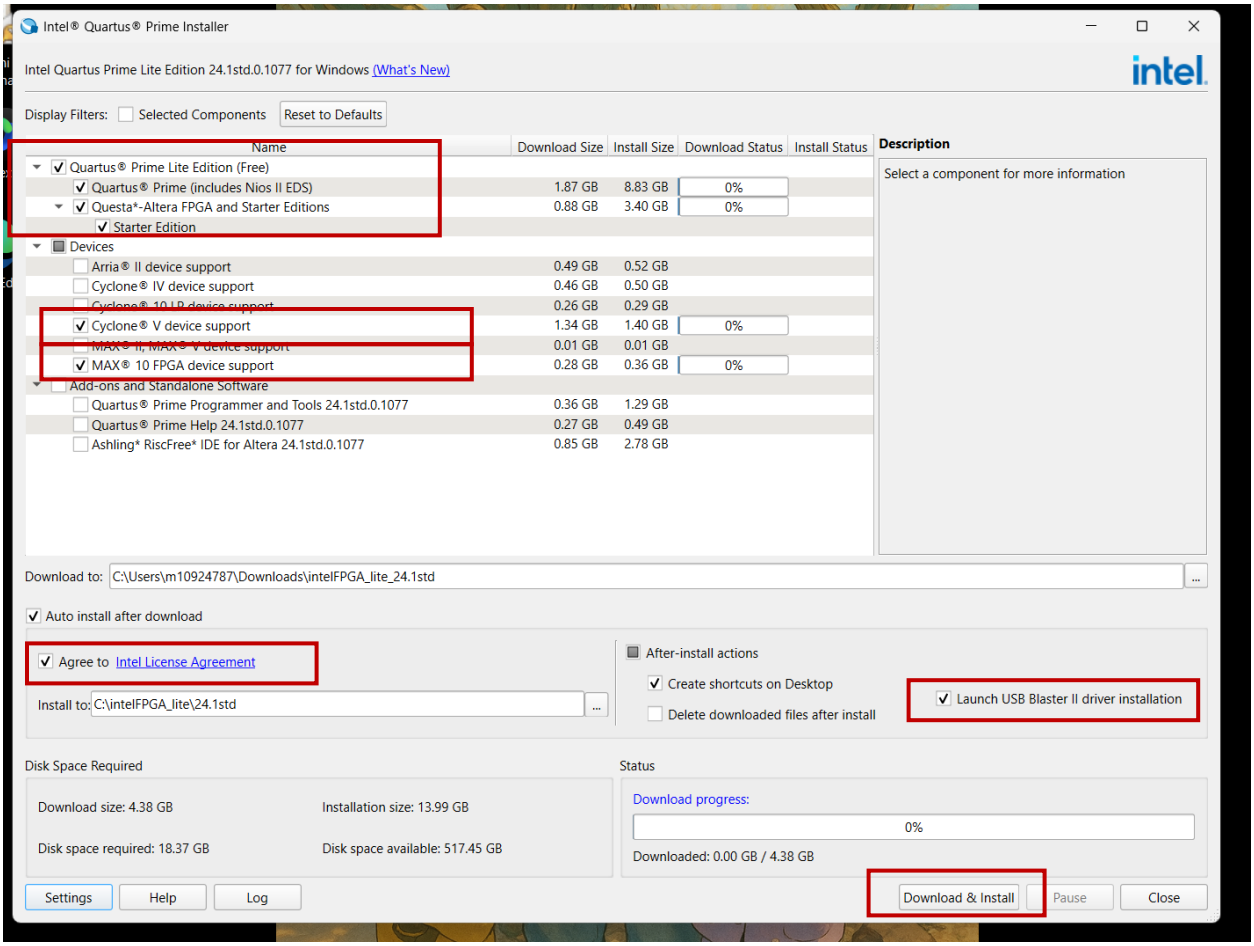
1. At the time of creating this document the latest version of Quartus Prime Lite for a Windows operating system is 24.1
2. Go to the link: https://www.intel.com/content/www/us/en/software-kit/849770/intel-quartus-prime-lite-edition-design-software-version-24-1-for-windows.html



3. Scroll down to 'Downloads' and download 'Intel® Quartus® Prime Installer (Recommended)'

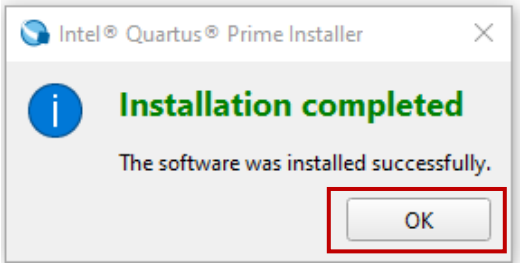


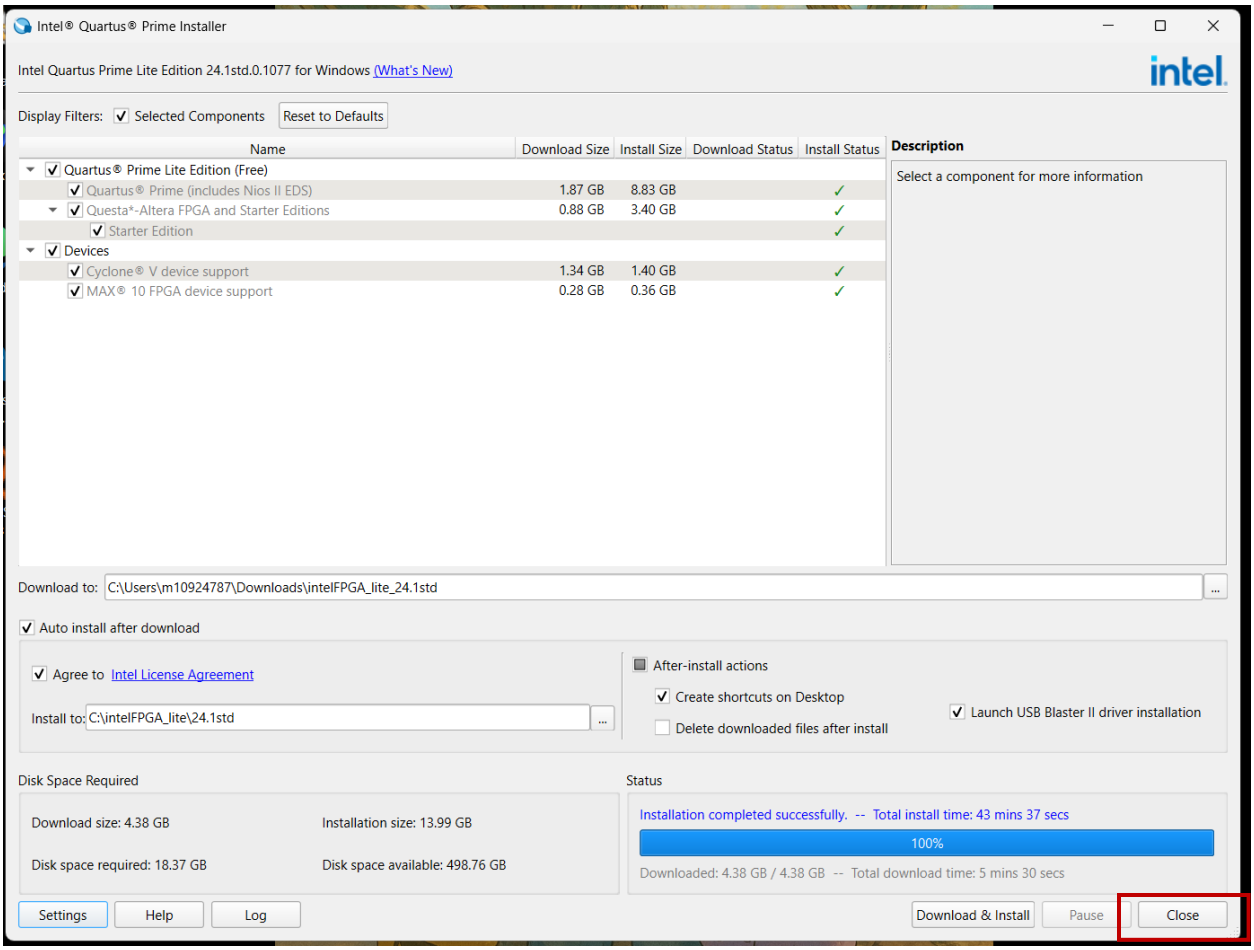
4. Save this somewhere accessible. Now double click on the installer application (.exe) that you just downloaded.



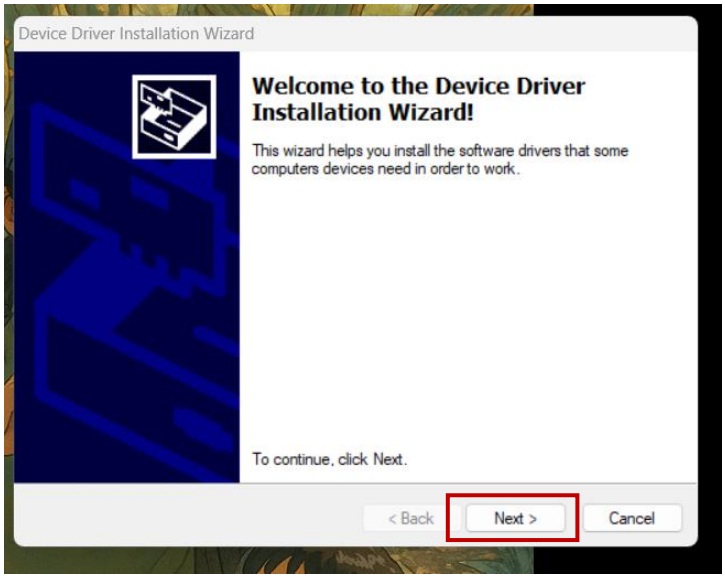
Get a cup of coffee and relax, it can take a while!

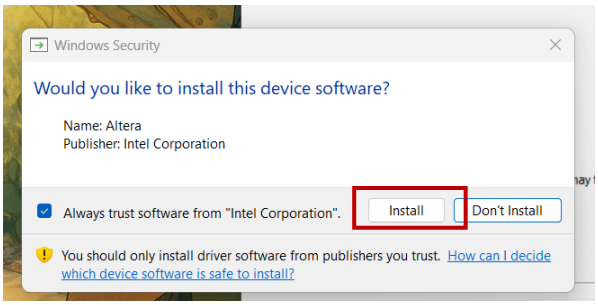
5. Here is what you will see once the download and installation are complete (took ~44 min):

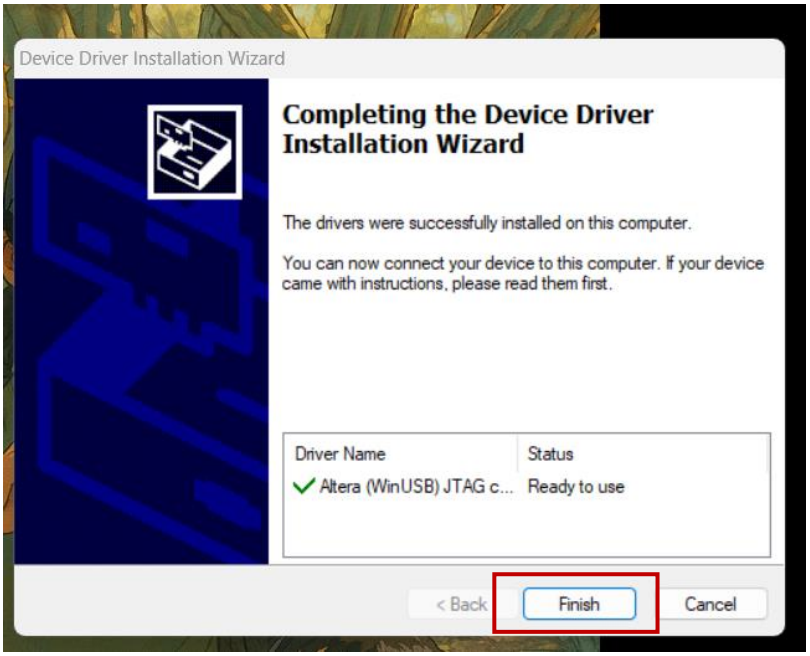




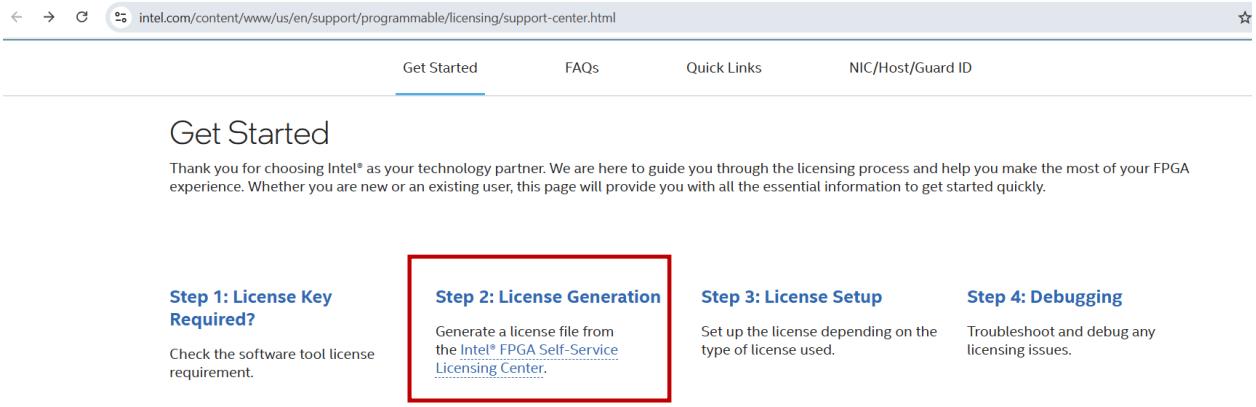
6. The device driver installation wizard will launch automatically. Follow the snapshots and the red rectangles below!



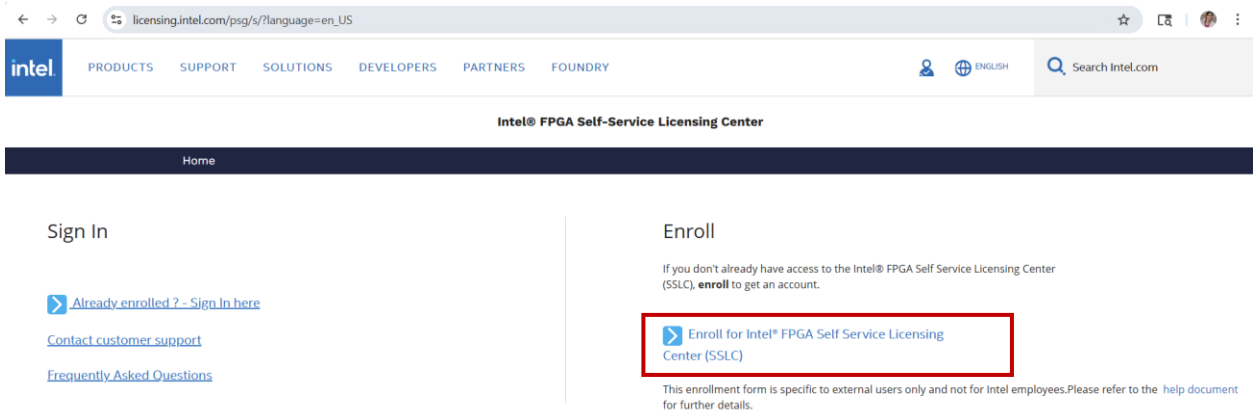




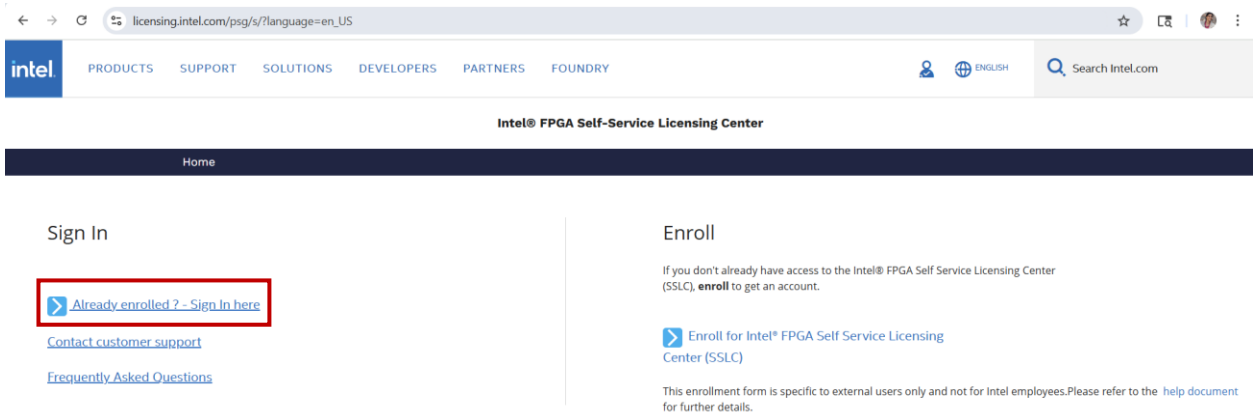
7. We will use a simulator 'Questa' with quartus. Using this simulator requires us to get a (free) license from intel. The following steps will guide you through the process. Go to intel's Self Service Licensing Center (SSLC) and click the link in Step 2: https://www.intel.com/content/www/us/en/support/programmable/licensing/support-center.html



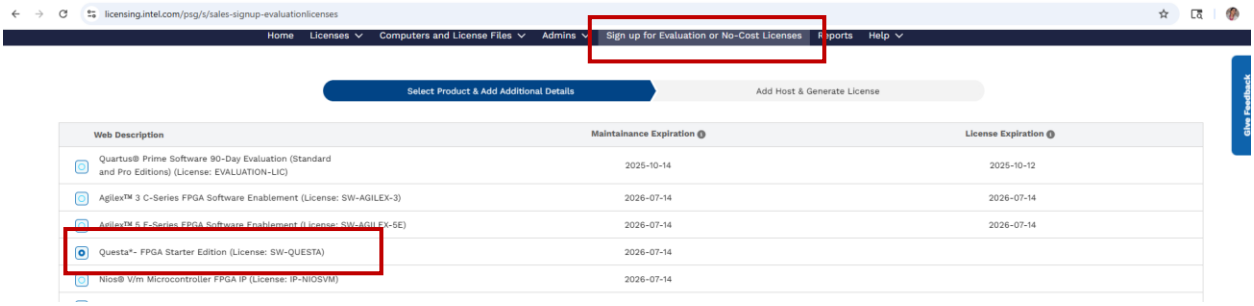
8. If you have used intel's SSLC before and have an account, simply login. If not, then enroll for the service. USE YOUR MINES EMAIL ID for this.



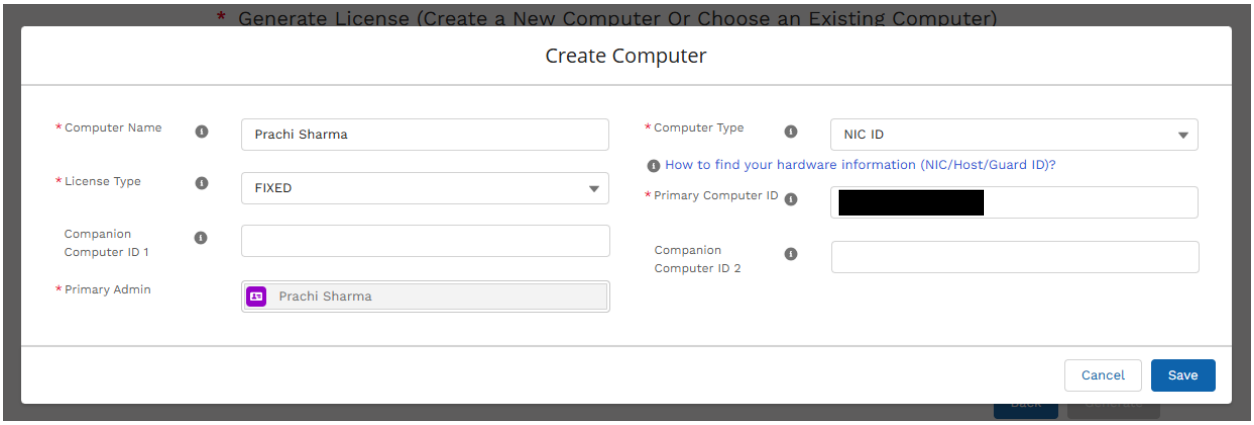
9. Enter your email and fill out the details requested in the next step to verify your email. Enter the verification code sent to your email and 'Create an account'
10. Login to your account using 'Sign in



11. Click on 'Sign up for Evaluation or No-Cost Licenses and choose Questa-FPGA Starter Edition (License: SW-QUESTA)

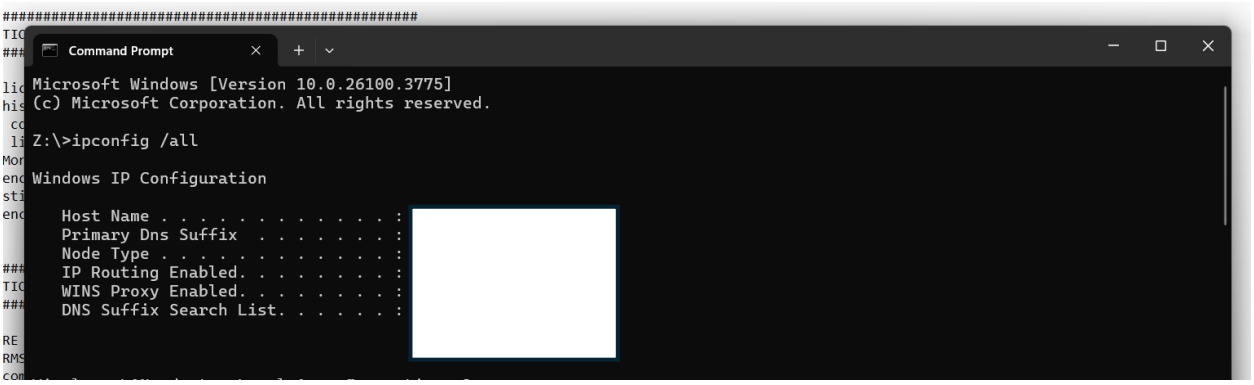


12. On the next page, ‘Create a new computer’ and fill in information as below. Your Computer name should be your own name (or any other name you want to use).

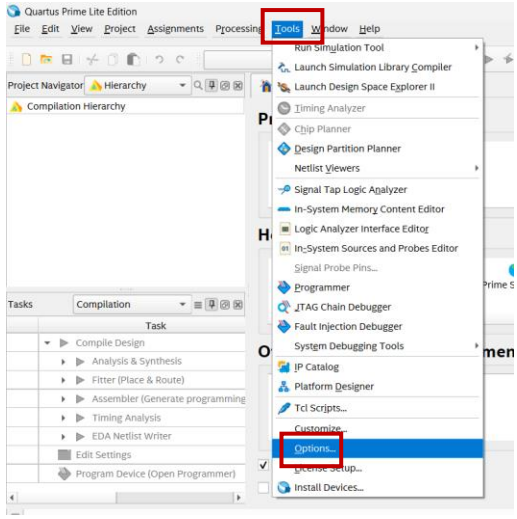


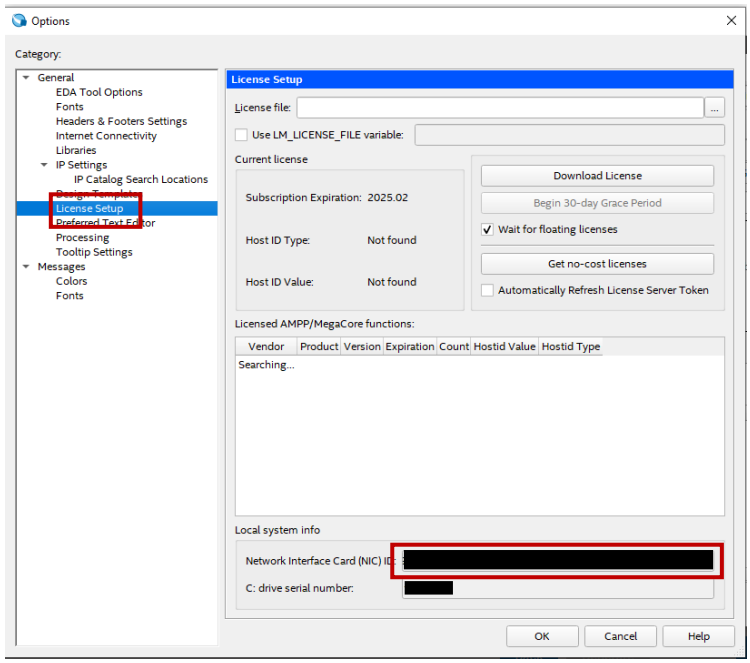
13. To find the NIC ID, we need to use two methods. YOU MUST CHOOSE A NIC ID COMMON BETWEEN BOTH STEPS a. AND b.
    a. Open the command prompt and type:
       >ipconfig /all

    You can see more than one result, pick one ‘Physical Address’. This is the NIC ID you will fill as Primary Computer ID. Don’t put a dash(-) between the alphabets/numbers of the Primary computer ID.

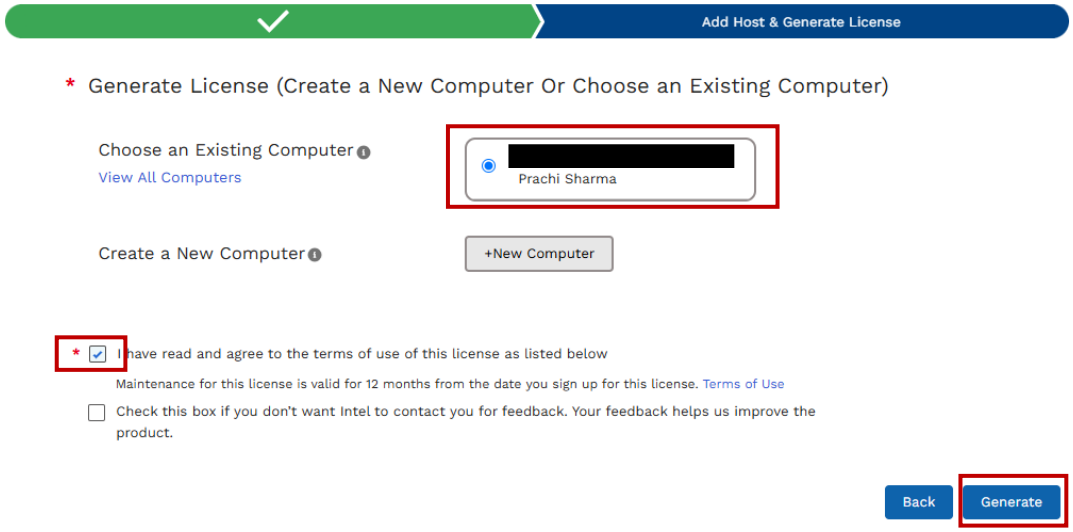


b. Launch Quartus. In the menu options in the toolbar at the top of the screen, choose ‘Options’ -> go to ‘License Setup’ and choose a NIC ID. Look at the screenshots below:

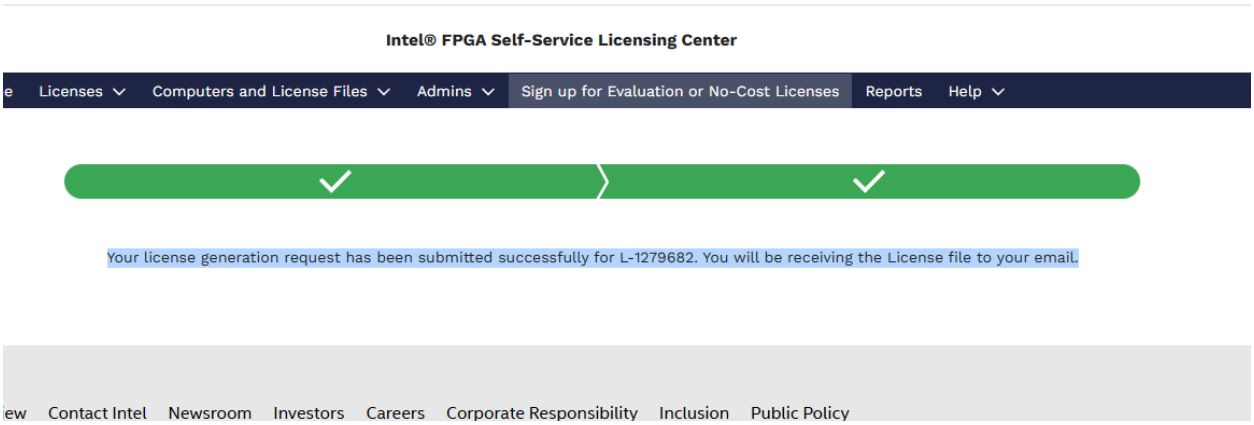




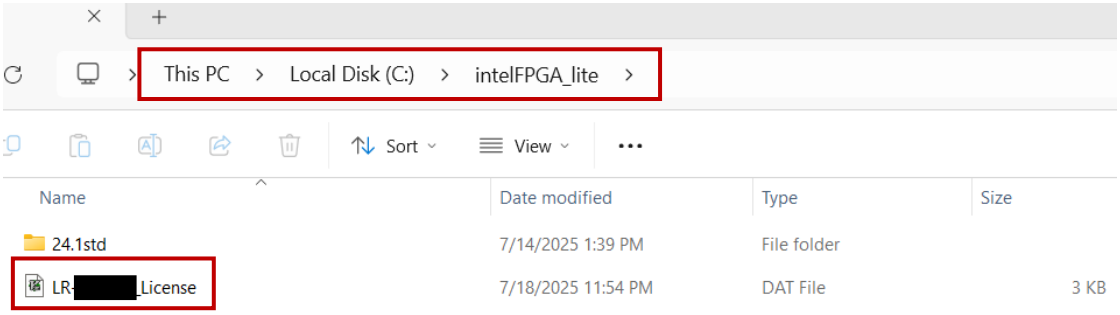
14. Fill the NIC ID in step 12. Once you have filled out all the information requested in step 12, you will see the computer you added showing up to generate a license. Agree to the terms of use and click ‘Generate’.



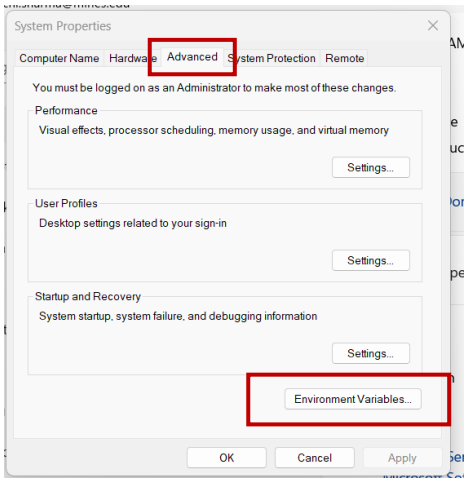
You should see the following screen. You will receive the license file on the email you used to enroll for SSLC.



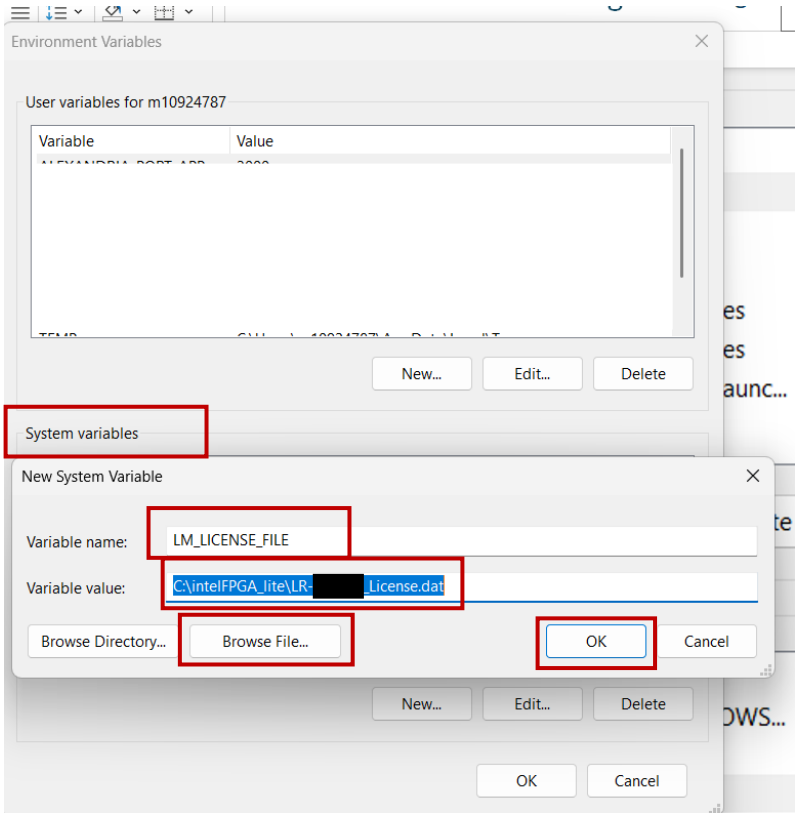
15. After getting the license file from intel on email, save it in some location where you won't delete it. I have saved in C:\intelFPGA_lite as LR-XXXXXX_License.dat



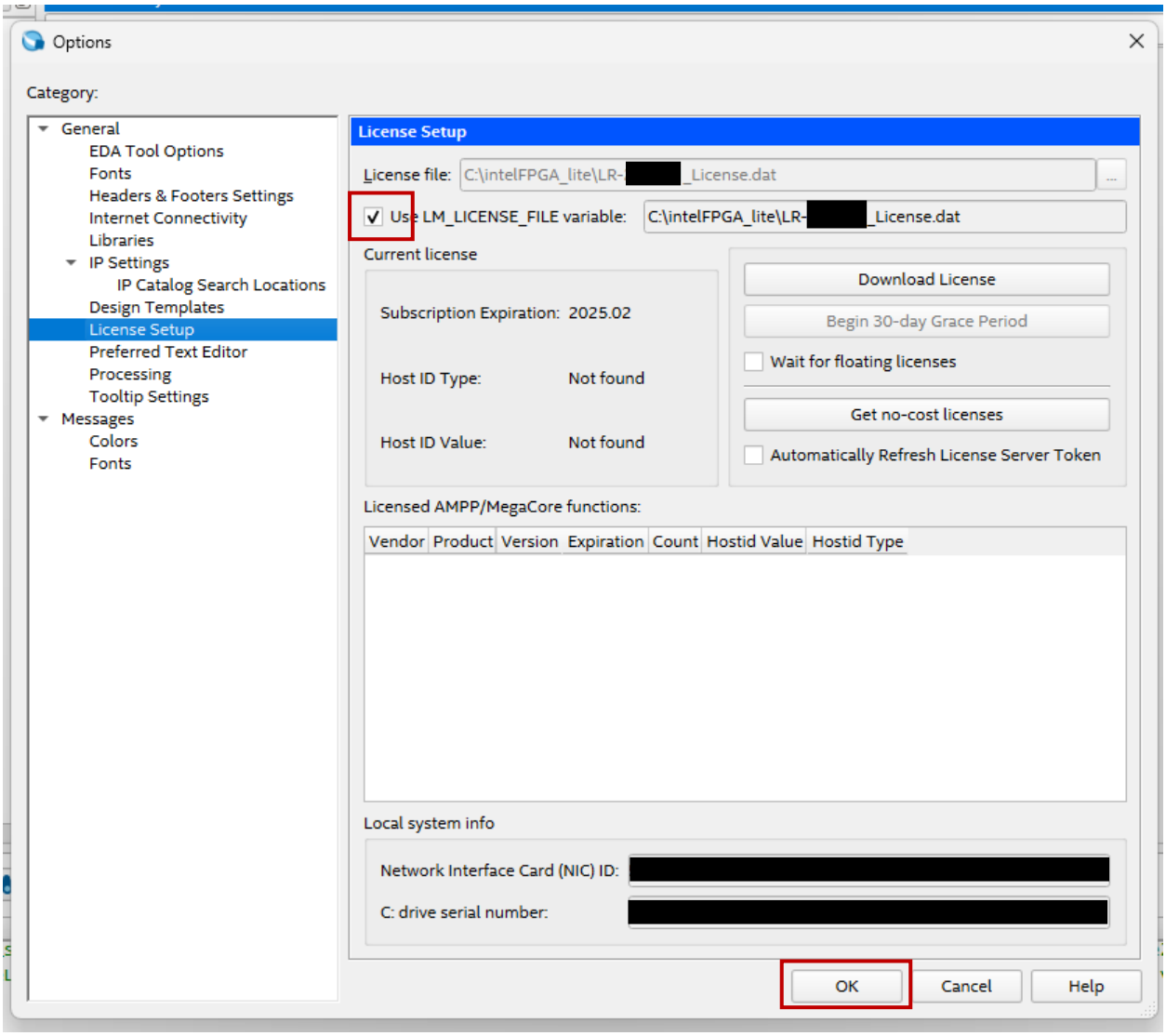
16. Now we will add environment variable to access the license file. Click on the windows start button and go to - > Settings -> System -> Advanced system settings-> Environment Variables.



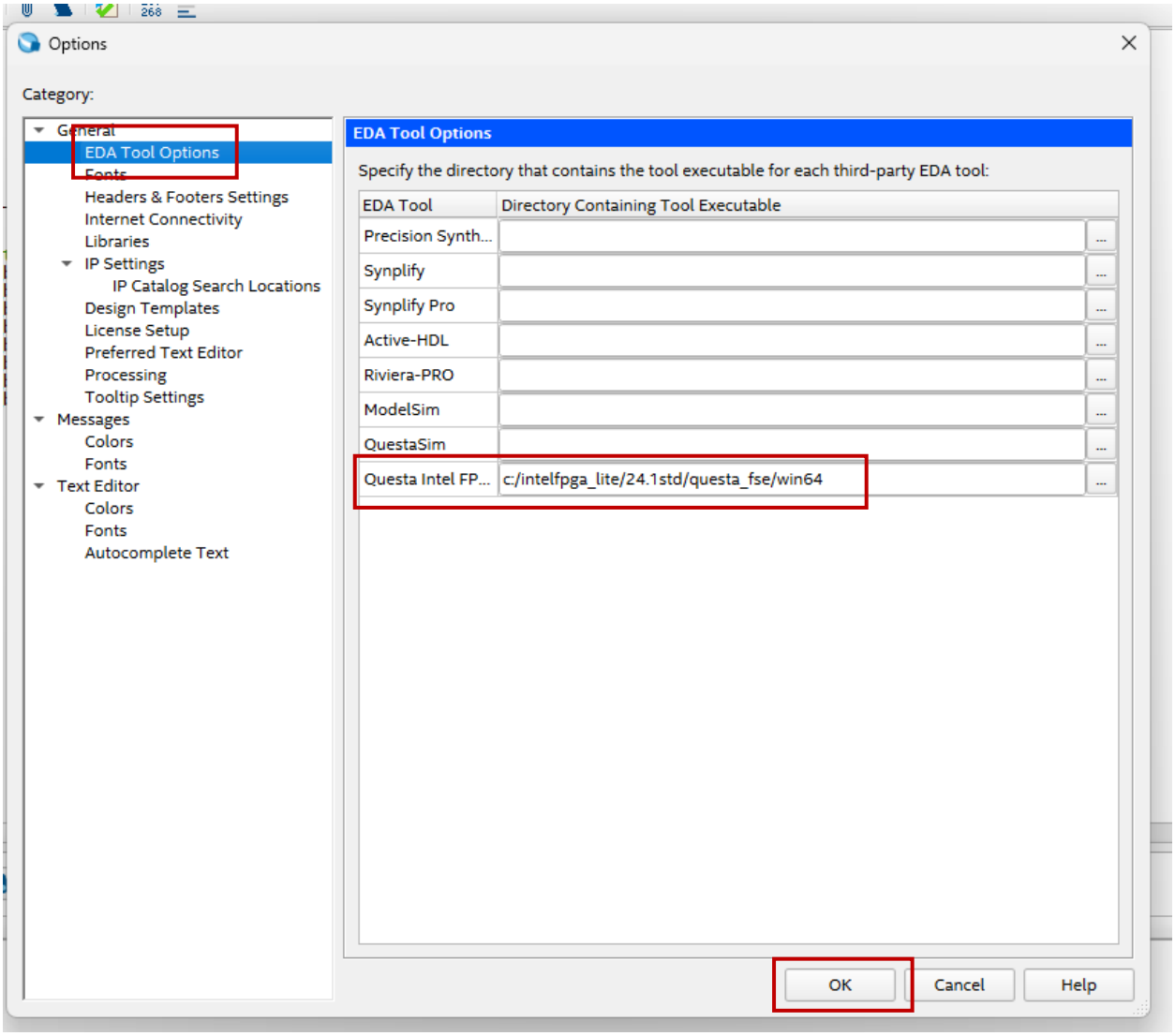
17. Under System variables -> New -> Variable name as LM_LICENSE_FILE and then 'Browse File' to the location where you have saved the license file from step 15.



18. Launch Quartus (if you don't already have it open!). Go to Tools -> Options -> License Setup. This is similar to step 13b. You will now see the License File is recognized. Check the box against 'Use LM_LICENSE_FILE variable'.



19. Go to Tools -> Options -> EDA Tool Options -> Questa Intel FPGA (last one on the list). Check to ensure the path is same as in the snapshot below:



20. Give yourself a pat on the back, you are all set to build your first project!